perfect cancellation when they cross, so the water at that location will not move at all.

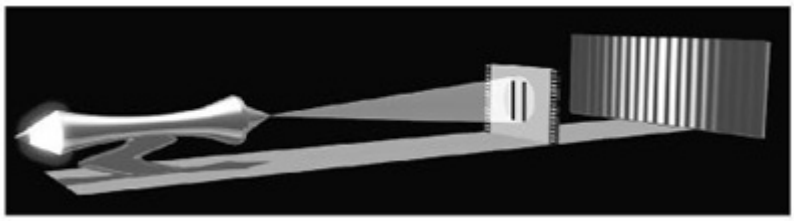


Figure 4.1 Laser light passing through two slits etched on a piece of black film yields an interference pattern on a detector screen, showing that light is a wave.

The same principle explains the pattern that light forms when it passes through the two slits in Figure 4.1. Light is an electromagnetic wave; when it passes through the two slits, it splits into two waves that head toward the screen. Like the two water waves just discussed, the two light waves interfere with each other. When they hit various points on the screen, sometimes both waves are at their peaks, making the screen bright; sometimes both waves are at their troughs, also making it bright; but sometimes one wave is at its peak and the other is at its trough and they cancel, making that point on the screen dark. We illustrate this in Figure 4.2b.

When the wave motion is analyzed in mathematical detail, including the cases of partial cancellations between waves at various stages between peaks and troughs, one can show that the bright and dark spots fill out the bands seen in Figure 4.1. The bright and dark bands are therefore a telltale sign that light is a wave, an issue that had been hotly debated ever since Newton claimed that light is not a wave but instead is made up of a stream of particles (more on this in a moment). Moreover, this analysis applies equally well to *any* kind of wave (light wave, water wave, sound wave, you name it) and thus, interference patterns provide the metaphorical smoking gun: you know you are dealing with a wave if, when it is forced to pass through two slits of the right size (determined by the distance between the wave's peaks and troughs), the resulting intensity pattern looks like that in Figure 4.1 (with bright regions representing high intensity and dark regions being low intensity).